COS2A: Conversion From Sentinel-2 to AVIRIS Hyperspectral Data Using Interpretable Algorithm With Spectral–Spatial Duality

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***Abstract*—The Sentinel-2 satellite, launched by the European Space Agency (ESA), offers extensive spatial coverage and has become indispensable in a wide range of remote sensing applications. However, it just has 12 spectral bands, making substances/objects identification less effective, not mentioning the varying spatial resolutions (10/20/60 m) across the 12 bands. If such a multiresolution 12-band image can be computationally converted into a hyperspectral image with uniformly high resolution (i.e., 10 m), it significantly facilitates remote identification tasks. Though there are some spectral super-resolution methods, they did not address the multiresolution issue on one hand, and, more seriously, they mostly focused on the CAVE-level hyperspectral image reconstruction (involving only 31 visible bands) on the other hand, greatly limiting their applicability in real-world remote sensing scenarios. We ambitiously aim to convert Sentinel-2 data directly into NASA's AVIRIS-level hyperspectral image (encompassing up to 172 visible and near-infrared (NIR) bands, after ignoring those absorption/corruption ones). For the first time, this article solves this specific super-resolution problem (highly ill-posed), allowing all historical Sentinel-2 data to have their corresponding high-standard AVIRIS counterparts. We achieve so by customizing a novel algorithm that introduces deep unfolding regularization and Q-quadratic-norm regularization into the so-called convex/deep (CODE) small-data learning criterion. Based on the derived spectral–spatial duality, the proposed interpretable COS2A algorithm demonstrates superior spectral super-resolution results across diverse land cover types, as validated through extensive experiments. Source codes: https://github.com/IHCLab/COS2A.**

***Index Terms*—AVIRIS image, computational imaging, convex/deep (CODE) learning, hyperspectral image, Sentinel-2 satellite, spectral super-resolution.**

I. Introduction

THE Sentinel-2 satellite is part of European Space Agency's (ESA) Copernicus Program, designed for Earth observation, especially for monitoring land and coastal areas. Sentinel-2 satellite offers wide spatial coverage and has become indispensable in various remote sensing applications, including agriculture/crop monitoring, forestry management, water quality observation, land use and land cover change detection, as well as natural disaster monitoring [1], [2]. Due to inherent hardware constraints of imaging sensors, coupled with practical considerations such as atmospheric absorption and the requirement for an adequate signal-to-noise ratio (SNR), the spatial resolution of Sentinel-2 imaging systems typically differs across spectral bands. Specifically, Sentinel-2 data include four 10-m bands, six 20-m bands, and two 60-m bands. Note that the Sentinel-2 image just has 12 spectral bands, making remote substances/objects identification less effective, not mentioning the varying spatial resolutions across the 12 bands. Thus, in order to employ the strong identifiability [3], [4] of hyperspectral imagery to facilitate the subsequent remote sensing missions, such as change detections [5], [6] and anomaly detections [7], [8], we are thinking about the possibility of computationally converting such a multiresolution Sentinel-2 data into hyperspectral data with uniformly high resolution (i.e., 10 m).

Though there are some spectral super-resolution methods [9], [10], they did not address the multiresolution issue on one hand, and, more seriously, they mostly focused on the Harvard/CAVE-level hyperspectral image reconstruction (involving only 31 visible bands) on the other hand [11], [12]. This greatly limits their practical applicability in real-world remote sensing scenarios, which often require near-infrared (NIR) bands to have higher atmospheric penetration, to have better material identifiability, as well as to have more effective vegetation observation (e.g., mangrove dynamics) [13], [14]. Therefore, this work ambitiously aims to convert Sentinel-2 data directly into NASA's AVIRIS-level hyperspectral image [15], which, after removing those absorption/corruption bands, encompasses up to 172 narrowly spaced spectral bands contiguously ranging from visible to NIR wavelengths (0.4–2.5 μm). For the first time, this article solves this specific super-resolution problem, termed as conversion from Sentinel-2 to AVIRIS (COS2A) problem. The COS2A technique will allow those historical Sentinel-2 data to have their corresponding high-standard AVIRIS counterpart data over the same scene.

Although COS2A is a highly ill-posed inverse problem, it is technically feasible. In fact, spectral super-resolution has been demonstrated to be feasible not only in the imaging software arena but also in the camera hardware implementation [16].

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Specifically, to meet the strict hardware volume specification in the miniaturized satellites [17], [18], the flat and nanoscale metamaterials [19] have been employed recently to design compact hyperspectral imaging meta-systems, for which a multispectral meta-image is first generated and then spectrally super-resolved to computationally obtain the target hyperspectral imaging results [16]. This shows the feasibility of spectral super-resolution even in such a challenging real-world scenario. Very recently, the quantum deep network (QUEEN) [20], [21] has also been applied to achieve spectral super-resolution for the highly challenging multispectral unmixing problem, which is an ill-posed and underdetermined source separation task [22]. Specifically, the QUEEN-based algorithm serves as a virtual quantum light-splitting prism to further split a multispectral band into more hyperspectral bands, thereby achieving the unmixing task through an overdetermined transform with quantum deep image prior (QDIP) [23]. Nevertheless, existing spectral super-resolution methods have limited spectral upsampling factors less than or just around ten [11], [16], [23], [24], making them not applicable to the COS2A problem.

Some related spectral super-resolution methods include the multistage spatial–spectral fusion network (MSFN) [25] and reparameterizing coordinate-preserving proximity spectral interaction network (RepCPSI) [26]. MSFN is a U-Net-like convolutional neural network (CNN) designed to learn the spatial correlation and spectral self-similarity (SS), while RepCPSI is a lightweight network utilizing multibranch convolutional structures during the training phase to facilitate the spatial–spectral feature extraction. In RepCPSI, a coordinate-preserving neighboring spectrum-aware attention mechanism is further designed to maintain spatial positional information while capturing local spectral dependencies in a lightweight manner. Considering the highly ill-posed nature of the target COS2A problem, it is natural to employ the low-rank property of hyperspectral data to mitigate the ill-posedness. For example, Dian et al. [11] integrate the canonical polyadic (CP) decomposition within a deep network and enhance global feature modeling through multidimensional attention. By decomposing high-dimensional features into learnable 1-D basis vectors, the method reduces both computational and storage complexities. For another example, Lin et al. [27] integrate the alternating direction method of multipliers (ADMM) and adaptive moment estimation (ADAM) to achieve outstanding spectral super-resolution results via the ADMM-Adam theory, thereby solving the problem under an elegant mathematical scheme. Nevertheless, all the above methods focus just on the 31-band hyperspectral reconstruction only in the visible spectral range and thus are not applicable for COS2A.

We should mention the seminal work [24], referred to as the multistage spectral-wise transformer (MST). Although it is also a 31-band method, it achieved remarkable performance at the time it was proposed and won the first place in the renowned NTIRE challenge. MST is a transformer-based framework that employs a spectral-wise attention mechanism by treating each spectral band as an individual token to capture long-range dependencies along the spectral dimension. The MST network adopts a multistage structure to progressively refine spectral details. We should also mention some good works that attempted to super-resolve more hyperspectral bands (up to 102 bands), including the spectral-cascade diffusion model (SCDM) [28] and the unfolding spatiospectral super-resolution network (US3RN) [29]. SCDM adopts a diffusion-based multistage spectral refinement by modeling the spectral probability distribution, while US3RN solves the super-resolution problem via deep unfolding after a tricky reformulation stage. Although they still do not reach the AVIRIS-level hyperspectral reconstruction, their mechanisms are well developed in our view.

After investigating all the above works, we conclude that AVIRIS-level spectral super-resolution is too challenging to be effectively solved using conventional inverse imaging theories, especially when the AVIRIS data need to be computationally obtained from multiresolution data, making the COS2A problem even challenging. Thus, we will employ the so-called convex/deep (CODE) small-data learning theory [30], composed of a convex phase and a deep phase, which was customized for solving challenging hyperspectral inverse imaging problems (e.g., restoration of NASA's highly damaged data), especially under the circumstances of data scarcity. Note that in COS2A, the training data collection is indeed a daunting and laborious task. We not only need to identify spatiotemporal overlapping trajectories between AVIRIS and Sentinel-2 sensors, thereby obtaining the training data pairs containing AVIRIS and Sentinel-2 images acquired over the same spatial scene at the same time, but also need to manually conduct several preprocessing stages before the data can be used to meaningfully learn the deep COS2A function (see Section III).

Algorithm 1 COS2A Spectral Super-Resolution Algorithm

1: **Given** Sentinel-2 image $\boldsymbol{Y}_S$.
2: Compute the deep rough solution $\boldsymbol{Y}_{\text{DE}}$ by Algorithm 2.
3: Compute the spectral response $\widetilde{\boldsymbol{D}}$ by ridge regression using $\boldsymbol{Y}_{\text{DE}}$ and (11).
4: Transform the COS2A problem (4) (with $\widetilde{\boldsymbol{D}}$ and $\boldsymbol{Y}_{\text{DE}}$) into its dual spatial super-resolution problem via Theorem 1.
5: Conduct the dual spatial super-resolution using fast convex algorithm [33] to obtain the solution $(\boldsymbol{A}^\star, \boldsymbol{S}^\star)$ of (4).
6: **Output** AVIRIS-level hyperspectral image $\boldsymbol{Y}_H^\star := \boldsymbol{A}^\star \boldsymbol{S}^\star$.

On the other hand, as we do not think the complicated mapping from Sentinel-2 to AVIRIS could be well learned merely using a deep learning (DE) network, we consider the deep solution just as an auxiliary regularization role. This greatly mitigates the burden of the deep phase in CODE learning and can be implemented based on the $\boldsymbol{Q}$-quadratic-norm regularization scheme in the CODE theory, which will be mathematically introduced in Section II-B. Thus, we will just have a rough deep solution, which, according to the CODE theory, will be fed into a convex optimization (CO) algorithm to complete the spectral super-resolution task. Again, the COS2A super-resolution is too challenging to be directly solved using conventional approaches, so we do not solve the CO phase directly. Instead, we derive a spectral–spatial duality theorem to transform the highly ill-posed COS2A spectral super-resolution problem into another well-known

and more well-studied spatial super-resolution problem (see Theorem 1), thereby elegantly solving COS2A via simple matrix factorizations. This is the so-called inverse problem transform [31], [32], allowing a hard inverse problem to be reformulated into and solved as a simpler and more well-developed inverse problem. Both our deep and convex phases are interpretable, and the induced COS2A algorithm (i.e., Algorithm 1) demonstrates superior spectral super-resolution results across diverse land cover types.

The remaining parts of this article will be organized as follows. In Section II, the COS2A problem is formulated and solved using CODE theory and spectral–spatial duality. Data preparation and extensive experimental results are presented in Section III. Conclusions are drawn in Section IV, and mathematical details are given in Appendix A.

II. Proposed COS2A Algorithm

A. Problem Description and Challenges

Given a multiresolution Sentinel-2 image $\boldsymbol{Y}_S \in \mathbb{R}^{12\times L}$, where L is the number of high-resolution pixels in the 10-m band, we aim to computationally reconstruct the corresponding AVIRIS-level hyperspectral image $\boldsymbol{Y}_H \in \mathbb{R}^{172\times L}$ over the same spatial scene ($M := 172$). The 12-band Sentinel-2 multiresolution image ($M_m := 12$) can be concisely represented as a matrix $\boldsymbol{Y}_S$ [34] because those pixels in the 20-m bands (resp., 60-m bands) are copied four times (resp., 36 times), as it is when downloaded from ESA's data portal. Also, although AVIRIS has 224 bands in its original hardware specification, a majority of the literature uses only those uncorrupted bands after removing those absorption/corruption ones, and the remaining 172-band high-quality hyperspectral data (termed as AVIRIS-like data) are proven to be more than sufficient in achieving challenging substances/objects remote identification tasks [35].

This inverse problem is highly ill-posed and regarded as more challenging than typical spectral super-resolution problems that often focus just on reconstructing the 31-band CAVE-level hyperspectral data [26]. Furthermore, it is very difficult to collect high-quality AVIRIS/Sentinel-2 data pairs acquired over the same spatial scene and at the same time, not mentioning the intrinsic noninteger multiple of their spatial resolutions and different flight trajectories/directions [15], [36]. This induces a small-data learning problem setting, thereby motivating us to adopt the CODE learning theory, which was originally developed/customized for restoring NASA's damaged hyperspectral image under the circumstances of data scarcity (and later on applied to numerous satellite monitoring missions) [14], [37]. Simply speaking, CODE avoids complicated mathematics and sophisticated DE procedure (e.g., big data collection or architecture design), by using $\boldsymbol{Q}$-quadratic-norm regularization to blend the advantages of two powerful inverse imaging techniques—CO and DE.

In Section II-B, we outline and recall the overall CODE learning procedure and formulate the problem of converting Sentinel-2 to AVIRIS data (i.e., COS2A problem) based on the CODE theory [30]. In Section II-C, we design an algorithm to solve the formulated COS2A problem. The implementation details of our COS2A algorithm are then collectively presented in Section II-D.

B. CODE-Driven COS2A Problem Formulation

Let us begin by recalling the CODE learning theory [30], which was originally proposed for economically solving remote sensing problems without requiring math-heavy optimization or big data collection. Simply speaking, CO often requires math-heavy regularization (but usually not requiring big data), and DE often relies on big data collection and daunting data labeling procedure (but usually not involving advanced mathematics). CODE learning theory blends the advantages from CO and DE, thereby solving inverse imaging problems without requiring big data or sophisticated regularization. CODE theory achieved so by bridging CO and DE using the so-called $\boldsymbol{Q}$-quadratic-norm regularizer $\|\cdot\|_{\boldsymbol{Q}}$, which not only has a very simple mathematic form but also is able to extract useful feature information learned from small data [30].

To be specific, given a target inverse imaging problem, the CODE theory proposes to adopt the following learning criterion to reconstruct the target image $\boldsymbol{Y}^\star$:

$$\boldsymbol{Y}^\star := \arg\min_{\boldsymbol{Y}\in\mathcal{Y}} \mathrm{DF}(\boldsymbol{Y} \mid \boldsymbol{X}) + \lambda \mathrm{REG}(\boldsymbol{Y}) \tag{1}$$

where $\mathcal{Y}$ often specifies some natural constraints to the target image, $\mathrm{DF}(\boldsymbol{Y} \mid \boldsymbol{X})$ is called the data-fitting term (or data fidelity term) given the observable quantity/data $\boldsymbol{X}$, and $\mathrm{REG}(\cdot) \triangleq (1/2)\|\boldsymbol{Y} - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2$ is employed as the regularization term with strength controlled by $\lambda \geq 0$. That said, a precomputed "deep" learning solution $\boldsymbol{Y}_{\mathrm{DE}}$ is plugged into the "convex" $\boldsymbol{Q}$-quadratic-norm function for computationally efficient regularization [30], where $\boldsymbol{Q}$ is a positive semidefinite (PSD) matrix with the induced norm of a given vector $\boldsymbol{v}$ defined as $\|\boldsymbol{v}\|_{\boldsymbol{Q}}^2 \triangleq \boldsymbol{v}^T\boldsymbol{Q}\boldsymbol{v}$ [38].

We have three remarks for the CODE regularization scheme, i.e., $\mathrm{REG}(\cdot) \triangleq (1/2)\|\boldsymbol{Y} - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2$. First, $\boldsymbol{Y}_{\mathrm{DE}}$ could be just a rough DE solution [30], meaning that neither a complicated deep network architecture nor a big training dataset is required, greatly mitigating the pressure in developing DE algorithms, especially when the target inverse problems are highly challenging (e.g., the COS2A problem). Second, compared to those commonly used total-variation (TV) regularization that is nondifferentiable or SS regularization that is math-heavy (and usually defined as a nonconvex function), the proposed $\boldsymbol{Q}$-quadratic-norm regularization is not only differentiable but also has a mathematically very simple form (a concisely defined convex function). Third, although $\|\cdot\|_{\boldsymbol{Q}}$ is simple, its regularization effectiveness is arguably stronger than those typical TV or SS regularizers [30, Footnote 1], as $\boldsymbol{Y}_{\mathrm{DE}}$ often possesses both the edge/similarity structural information embedded in TV/SS (and such information can be extracted by $\|\cdot\|_{\boldsymbol{Q}}$ and fused into the final solution $\boldsymbol{Y}^\star$). This has been both theoretically and experimentally verified in [30], thereby extensively inducing numerous successful CODE-based remote sensing applications [16], [39].

Accordingly, let us start formulating the COS2A problem based on the CODE theory. If the spectral response $\boldsymbol{D}$ between

Sentinel-2 and AVIRIS spectral bands is available, it is natural to define the data fidelity term as $\mathrm{DF}(\boldsymbol{Y}_H \mid \boldsymbol{Y}_S) = \|\boldsymbol{Y}_S - \boldsymbol{D}\boldsymbol{Y}_H\|_F^2$. However, we did not identify an officially released spectral response specification, and hence, the matrix $\boldsymbol{D}$ should be estimated. To facilitate the estimation, we focus only on the four highest resolution bands (i.e., 10-m bands) of the Sentinel-2 image [see (11)], yielding a reliable spectral response matrix $\widetilde{\boldsymbol{D}} \in \mathbb{R}^{4\times 172}$ that will be proven to be sufficient for effectively solving COS2A, and hence, the data fidelity term will be designed as $\mathrm{DF}(\boldsymbol{Y}_H \mid \widetilde{\boldsymbol{Y}}_S) \triangleq \|\widetilde{\boldsymbol{Y}}_S - \widetilde{\boldsymbol{D}}\boldsymbol{Y}_H\|_F^2$, in which $\widetilde{\boldsymbol{Y}}_S \in \mathbb{R}^{4\times L}$ represents the concatenation of the four 10-m Sentinel-2 bands. The estimation criterion of $\widetilde{\boldsymbol{D}}$ will be presented in Section II-D. Thus, to complete the design of the CODE criterion (1) for COS2A, the remaining tasks are twofold—the design of the regularization term and the design of the constraint set $\mathcal{Y}$.

First, the regularization form is defined as those in typical CODE criteria [30], namely $\mathrm{REG}(\boldsymbol{Y}_H) \triangleq (1/2)\|\boldsymbol{Y}_H - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2$, where the deep rough solution $\boldsymbol{Y}_{\mathrm{DE}}$ will be obtained using simple deep unfolding network, as will be detailed in Section II-D. As for the PSD matrix $\boldsymbol{Q}$ in the regularizer, it will be customized for $\boldsymbol{Y}_{\mathrm{DE}}$ to compensate the insufficiency of $\boldsymbol{Y}_{\mathrm{DE}}$, as will also be detailed in Section II-D. Thus, (1) leads to the following COS2A formulation:

$$\boldsymbol{Y}_H^\star := \arg\min_{\boldsymbol{Y}_H\in\mathcal{Y}} \|\widetilde{\boldsymbol{Y}}_S - \widetilde{\boldsymbol{D}}\boldsymbol{Y}_H\|_F^2 + \frac{\lambda}{2}\|\boldsymbol{Y}_H - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 \tag{2}$$

where we set $\lambda := 2$ in this work. The norm $\|\cdot\|_{\boldsymbol{Q}}$ has been previously defined below (1) for the vector input, and for the matrix input, the matrix is first vectorized to fit the standard definition [38]. Second, to specify the constraint set $\mathcal{Y}$, we need to discuss the desired nature of $\boldsymbol{Y}_H$. The probably most frequently noticed one would be the low-rank nature of typical hyperspectral image $\boldsymbol{Y}_H = \boldsymbol{A}\boldsymbol{S}$, where $\boldsymbol{A} \in \mathbb{R}^{172\times N}$ and $\boldsymbol{S} \in \mathbb{R}^{N\times L}$ are often considered as the hyperspectral endmember matrix and the abundance matrix, respectively, in which N denotes the matrix rank (often much less than 172 and L) [40], [41], [42]. Hierarchically, driven by the low-rank nature of $\boldsymbol{Y}_H = \boldsymbol{A}\boldsymbol{S}$, we further employ the minimum-volume nature of $\boldsymbol{A}$ [43] and the sparsity nature of $\boldsymbol{S}$ [44] to implicitly define $\mathcal{Y}$ at this moment, leading to the COS2A criterion, i.e.,

$$\boldsymbol{Y}_H^\star := \arg\min_{\boldsymbol{Y}_H=\boldsymbol{A}\boldsymbol{S}\in\mathcal{Y}} \|\widetilde{\boldsymbol{Y}}_S - \widetilde{\boldsymbol{D}}\boldsymbol{A}\boldsymbol{S}\|_F^2 + \frac{\lambda}{2}\|\boldsymbol{A}\boldsymbol{S} - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2. \tag{3}$$

We will discuss more theoretical aspects about the volume and sparsity constraints in Section II-C, thereby making $\mathcal{Y}$ explicit, and accordingly derive the COS2A algorithm for solving (3).

C. Algorithm for Solving the COS2A Problem

Before solving the COS2A problem (3), we first discuss the theoretical foundation of the volume constraint for $\boldsymbol{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N]$, where $\boldsymbol{a}_i$ is the hyperspectral signature (endmember) of the ith underlying substance presented in $\boldsymbol{Y}_H$. According to the well known Craig's criterion [45], [46] in remote sensing area, the high-fidelity endmembers can often be estimated by the vertices of the minimum-volume simplex that encloses the pixel vectors, and under some mild conditions, such an estimation can be proven to yield perfect endmember estimation [4], [47]. Therefore, we can add an additional simplex volume term $\mathrm{vol}(\boldsymbol{A})$ in the objective function to explicitly promote the minimum-volume constraint. However, simplex volume is known to be nonconvex [38], [48], motivating us to adopt its convex surrogate, which is defined as $\mathrm{vol}(\boldsymbol{A}) \triangleq (1/2)\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\|\boldsymbol{a}_i - \boldsymbol{a}_j\|_2^2$.

On the other hand, the sparsity constraint for $\boldsymbol{S}$ is oriented from the abundance sparsity nature [49], motivating us to add another explicit ℓ_1-norm term $\|\boldsymbol{S}\|_1$ in the objective function [38]. All in all, criterion (3) can be explicitly written as

$$\begin{aligned}\min_{\boldsymbol{A},\boldsymbol{S}\geq\boldsymbol{0}} \|\widetilde{\boldsymbol{Y}}_S - \widetilde{\boldsymbol{D}}\boldsymbol{A}\boldsymbol{S}\|_F^2 + \frac{\lambda}{2}\|\boldsymbol{A}\boldsymbol{S} - \boldsymbol{Y}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 \\ + \alpha\mathrm{vol}(\boldsymbol{A}) + \beta\|\boldsymbol{S}\|_1\end{aligned} \tag{4}$$

where we further add the nonnegative constraints for both endmembers and abundances, and $\alpha, \beta \geq 0$ can be considered as penalty parameters to penalize the violation of the natures discussed in Section II-B. We empirically set $\alpha = \beta = 0.002$. Once we obtain the optimal results $(\boldsymbol{A}^\star, \boldsymbol{S}^\star)$ from the explicit COS2A criterion (4), we will be able to reconstruct the target image as $\boldsymbol{Y}_H^\star = \boldsymbol{A}^\star\boldsymbol{S}^\star$.

Although we can design an alternating optimization algorithm to solve (4), which amounts to two large-scale convex subproblems for solving $\boldsymbol{A}$ and $\boldsymbol{S}$, we are thinking about solving (4) in a more elegant manner, such as by employing some more well-developed super-resolution theory (note that spectral super-resolution is less investigated in literature). Specifically, as spatial super-resolution methods have been extensively studied in remote sensing and computer vision areas, related theories and methods are considered highly effective [50]. Therefore, we are considering to transform the COS2A spectral super-resolution problem into its dual version called spatial super-resolution [50], thereby allowing us to elegantly solve the COS2A problem (4) by calling effective spatial super-resolution algorithms. This idea is inspired by the so-called inverse problem transform [31], [32]—simply speaking, if a tough problem (e.g., convolution) is well transformed into another space (e.g., Fourier-transformed space), it could dramatically become a much simpler problem (e.g., multiplication) that can be elegantly and efficiently solved.

According to the above philosophy, we derive the following theorem based on the nonnegative matrix factorization (NMF) theory [51], allowing us to judiciously solve the COS2A problem using well-developed spatial super-resolution theory.

Theorem 1: (The Spectral–Spatial Duality) The considered COS2A spectral super-resolution problem (4) is mathematically equivalent to the well-known coupled-NMF spatial super-resolution problem in optical remote sensing.□

To facilitate the readability of the main text, the proof of Theorem 1 will be given in Appendix A, where one can also find the explicit mathematical definition of the coupled-NMF spatial super-resolution problem commonly seen in the remote sensing area. Based on Theorem 1, we will solve (4) by calling a fast convex algorithm [33] that was customized for the coupled-NMF spatial super-resolution problem. The overall COS2A algorithm is graphically illustrated in Fig. 1 and summarized in Algorithm 1, whose implementation details will be given in Section II-D.

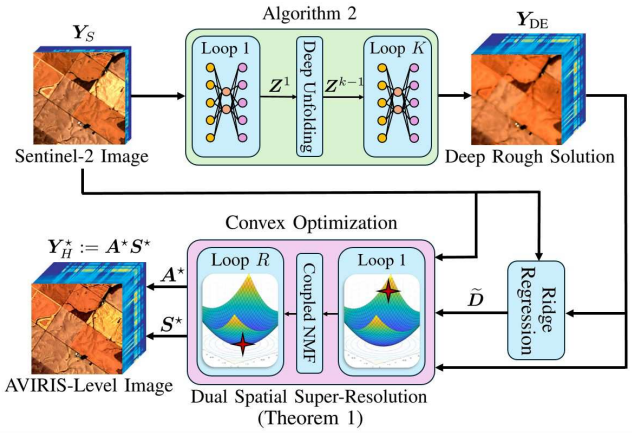


Fig. 1. Proposed COS2A spectral super-resolution algorithm (i.e., Algorithm 1) upgrades the Sentinel-2 multispectral image to AVIRIS-level hyperspectral image. It first computes the rough solution $\boldsymbol{Y}_{\text{DE}}$ using simple DE (i.e., Algorithm 2) and the scene-adaptive spectral response $\widetilde{\boldsymbol{D}}$ using ridge regression (11), as a preparation of the subsequent dual spatial super-resolution (see Theorem 1). The dual spatial super-resolution is a coupled-NMF problem and is solved using a fast convex algorithm [33], which returns the reconstructed endmember matrix $\boldsymbol{A}^\star$ and abundance matrix $\boldsymbol{S}^\star$. Overall, COS2A judiciously employs the CODE theory [30] to computationally obtain the AVIRIS-level image $\boldsymbol{Y}_H^\star := \boldsymbol{A}^\star \boldsymbol{S}^\star$ with high-fidelity spatial–spectral textures.

The computational time of the proposed CODE-based COS2A algorithm is dominated by the CO part for implementing the spectral–spatial duality, which is executed by coupled-NMF according to Theorem 1. The computational complexity has been analyzed in Appendix B. Specifically, the complexity is quadratic to the number of spectral bands M in practice and linear to the number of pixels L, as detailed in Appendix B. Note that the image size L often affects the computational time most significantly. Thus, having a complexity linear in L is a highly desired property, as reflected in some seminal image processing algorithms [52].

D. Algorithm Derivation and Implementation Details

Remaining tasks to complete the design of the COS2A algorithm are to explicitly construct the $\boldsymbol{Q}$-quadratic-norm regularization, to build a deep network to compute $\boldsymbol{Y}_{\text{DE}}$, as well as to estimate the spectral response function via $\boldsymbol{Y}_{\text{DE}}$. These implementation tasks will be done in this section.

Algorithm 2 Deep Unfolding Algorithm for Solving (6)

Given Sentinel-2 image $\boldsymbol{Y}_S$, and $\rho > 0$.
Initialize $k := 0$ and $\boldsymbol{U}^0 := \boldsymbol{0}$, and spectrally upsample $\boldsymbol{Y}_S$ to initialize $\boldsymbol{Y}_H^0$ (see Fig. 2).
repeat
  Update $\boldsymbol{Z}^{k+1} \in \arg\min_{\boldsymbol{Z}} \ \mathcal{L}(\boldsymbol{Y}_H^k, \boldsymbol{Z}, \boldsymbol{U}^k)$;
  Update $\boldsymbol{Y}_H^{k+1} \in \arg\min_{\boldsymbol{Y}_H} \ \mathcal{L}(\boldsymbol{Y}_H, \boldsymbol{Z}^{k+1}, \boldsymbol{U}^k)$;
  Update $\boldsymbol{U}^{k+1} := \boldsymbol{U}^k - \boldsymbol{Y}_H^{k+1} + \boldsymbol{Z}^{k+1}$;
  $k := k + 1$;
until the predefined stopping criterion is met.
Output $\boldsymbol{Y}_{\text{DE}} := \boldsymbol{Z}^k$.

In Algorithm 1, we need to compute a deep solution $\boldsymbol{Y}_{\text{DE}}$. According to the CODE theory [30], the deep solution $\boldsymbol{Y}_{\text{DE}}$ just needs to be a simple rough solution (thereby avoiding big data or complicated deep network), so we will design a simple deep unfolding network to compute $\boldsymbol{Y}_{\text{DE}}$ based on the following simple criterion:

$$\boldsymbol{Y}_{\text{DE}} := \arg\min_{\boldsymbol{Y}_H} \|\boldsymbol{Y}_S - \boldsymbol{D}\boldsymbol{Y}_H\|_F^2 + \text{DIP}(\boldsymbol{Y}_H) \tag{5}$$

where those sophisticated regularization schemes are all merged into the implicit deep regularizer DIP$(\cdot)$, which is theoretically valid based on the seminal deep image prior (DIP) theory [23], [53]. The implicit form greatly simplifies the deep network design after unfolding, well aligning with the philosophy of the CODE theory [30]. By the simplified DIP criterion (5), the associated COS2A deep network will be implemented to convert the Sentinel-2 image $\boldsymbol{Y}_S$ to the rough AVIRIS-level deep image $\boldsymbol{Y}_{\text{DE}}$ (see Algorithm 2).

The implicitly regularized simple COS2A criterion (5) can be deeply unfolded using the ADMM optimizer [38], thereby implementing the COS2A task using an explicitly constructed deep network. Specifically, by introducing an auxiliary variable $\boldsymbol{Z} \in \mathbb{R}^{M \times L}$, we first reformulate criterion (5) into the standard ADMM form, i.e.,

$$\min_{\boldsymbol{Y}_H = \boldsymbol{Z}} \|\boldsymbol{Y}_S - \boldsymbol{D}\boldsymbol{Y}_H\|_F^2 + \text{DIP}(\boldsymbol{Z}). \tag{6}$$

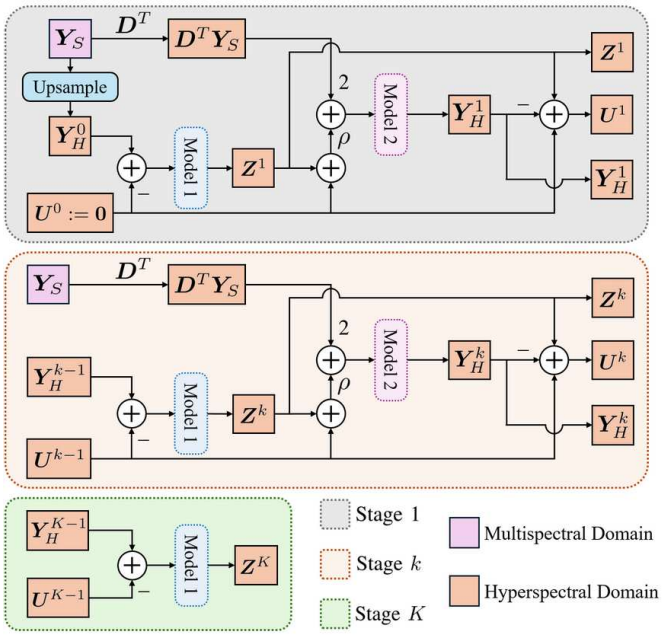


Fig. 2. To deeply unfold the K-stage Algorithm 2, our proposed deep architecture implements the operators in (8) and (9) using unfolded models 1 and 2 (see Fig. 3), respectively. This induces a very lightweight architecture (about 0.75 M parameters) implemented based on the group convolution. Simply speaking, the kth stage ($1 < k < K$) prepares two images [i.e., "$\boldsymbol{Y}_H^k - \boldsymbol{U}^k$" [see (8)] and "$2\boldsymbol{D}^T\boldsymbol{Y}_S + \rho(\boldsymbol{Z}^{k+1} + \boldsymbol{U}^k)$" [see (9)]] to be processed using model 1 [see (8)] and model 2 [see (9)], respectively, where ρ is a trainable parameter shared in each stage. The dual variable $\boldsymbol{U}^k$ is also updated according to Algorithm 2. Stage 1 further implements the initialization of Algorithm 2, while the last stage (i.e., Stage K) is simplified as only the target image $\boldsymbol{Z}^K$ is needed as the output of Algorithm 2. As measured in the floating-point operation (FLOP), the computational complexities when super-resolving a typical 256 $\times$ 256 image are 97.30G FLOPs for the initial stage (i.e., Stage 1), 96.87G FLOPs for the intermediate stage (i.e., Stage k with $1 < k < K$), and 96.15G FLOPs for the final stage (i.e., Stage K).

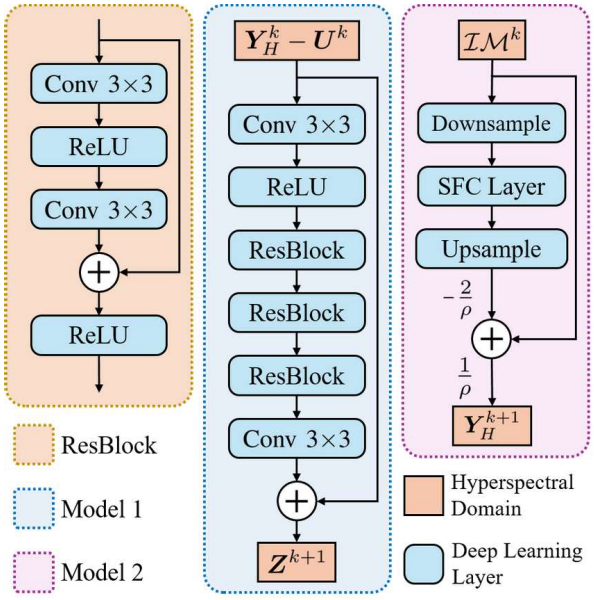


Fig. 3. Model 1 implements the proximal denoiser of (8) using the DIP-driven residual-in-residual deep convolutional (Conv) architecture. Model 2 implements the operator of (9) based on the symmetricity of $\boldsymbol{\Phi}$ and the physical meanings of $\boldsymbol{D}$ (downsampling) and $\boldsymbol{D}^T$ (upsampling). The two models serve as the basic units to deeply unfold the Algorithm 2.

The augmented Lagrangian of (6) [38] in the ADMM theory is defined as

$$\mathcal{L}_\rho(\boldsymbol{Y}_H, \boldsymbol{Z}, \boldsymbol{U}) = \|\boldsymbol{Y}_S - \boldsymbol{D}\boldsymbol{Y}_H\|_F^2 + \mathrm{DIP}(\boldsymbol{Z}) + \frac{\rho}{2}\|\boldsymbol{Y}_H - \boldsymbol{Z} - \boldsymbol{U}\|_F^2 \tag{7}$$

where $\boldsymbol{U} \in \mathbb{R}^{M\times L}$ is the scaled dual variable, $\rho > 0$ is referred to as the penalty parameter in ADMM, and ρ is set as a trainable parameter in the deep unfolding network (see Fig. 2). Then, the ADMM algorithm [38] for solving (6) via deep unfolding is summarized in Algorithm 2.

To complete Algorithm 2, we still need to implement the two optimization subproblems for $\boldsymbol{Z}^{k+1}$ and $\boldsymbol{Y}_H^{k+1}$, from which we can deploy the COS2A-driven deep unfolding network. By the definition of (7), the update of $\boldsymbol{Z}$ actually corresponds to the following proximal operator:

$$\boldsymbol{Z}^{k+1} := \mathrm{prox}_{\frac{1}{\rho}\mathrm{DIP}}\left(\boldsymbol{Y}_H^k - \boldsymbol{U}^k\right) \tag{8}$$

where $\mathrm{prox}_f(\boldsymbol{v}) \triangleq \arg\min_{\boldsymbol{x}} f(\boldsymbol{x}) + (1/2)\|\boldsymbol{x} - \boldsymbol{v}\|_2^2$ denotes the proximal operator associated with a given function f [54]. According to this definition of proximal operator $\mathrm{prox}_f(\boldsymbol{v})$ [54], it is nothing but a regularized denoising problem. Specifically, $\boldsymbol{v}$ is considered as a noisy image, from which we aim to recover a clean image $\boldsymbol{x}^\star \equiv \mathrm{prox}_f(\boldsymbol{v})$ with some desired property specified and promoted by the regularizer f. Therefore, the auxiliary hyperspectral image $\boldsymbol{Z}^{k+1}$ can be obtained by denoising the image "$\boldsymbol{Y}_H^k - \boldsymbol{U}^k$" using the deep network regularizer/prior (i.e., DIP) [53], [55]. Accordingly, the deep unfolding network for (8) can be deployed as a denoising network, for which the residual-in-residual deep architecture is known to be a simple but effective one [56], as detailed in model 1 of Fig. 3. The noisy image "$\boldsymbol{Y}_H^k - \boldsymbol{U}^k$" can be formed by the architecture deployed in Fig. 2 and then be fed into model 1 to implement (8).

On the other hand, by the definition of (7), the update of $\boldsymbol{Y}_H$ actually amounts to an unconstrained convex quadratic problem having the following solution:

$$\begin{aligned}\boldsymbol{Y}_H^{k+1} &= \arg\min_{\boldsymbol{Y}_H} \|\boldsymbol{Y}_S - \boldsymbol{D}\boldsymbol{Y}_H\|_F^2 + \frac{\rho}{2}\|\boldsymbol{Y}_H - \boldsymbol{Z}^{k+1} - \boldsymbol{U}^k\|_F^2 \\ &= \left(2\boldsymbol{D}^T\boldsymbol{D} + \rho\boldsymbol{I}\right)^{-1}\left(2\boldsymbol{D}^T\boldsymbol{Y}_S + \rho\left(\boldsymbol{Z}^{k+1} + \boldsymbol{U}^k\right)\right).\end{aligned}$$

As $\boldsymbol{D}$ is a fat matrix, the matrix inversion part in the deep unfolding network is not easily trainable. To resolve the dilemma, the Woodbury matrix inversion lemma [38] is applied to provide an equivalent solution, i.e.,

$$\boldsymbol{Y}_H^{k+1} := \frac{1}{\rho}\left(\boldsymbol{I} - \frac{2}{\rho}\boldsymbol{D}^T\boldsymbol{\Phi}\boldsymbol{D}\right)\left(2\boldsymbol{D}^T\boldsymbol{Y}_S + \rho\left(\boldsymbol{Z}^{k+1} + \boldsymbol{U}^k\right)\right) \tag{9}$$

where $\boldsymbol{\Phi} \triangleq (\boldsymbol{I} + (2/\rho)\boldsymbol{D}\boldsymbol{D}^T)^{-1}$ is a symmetric matrix just involving a small-scale matrix inversion. The input image "$\mathcal{IM}^k \triangleq 2\boldsymbol{D}^T\boldsymbol{Y}_S + \rho(\boldsymbol{Z}^{k+1} + \boldsymbol{U}^k)$" of (9) can be formed by the architecture deployed in Fig. 2 and then be fed into model 2 that implements the operator "$(1/\rho)(\boldsymbol{I} - (2/\rho)\boldsymbol{D}^T\boldsymbol{\Phi}\boldsymbol{D})$" in (9). Specifically, the identity matrix in the operator is implemented by a shortcut connection, and "$\boldsymbol{D}^T\boldsymbol{\Phi}\boldsymbol{D}$" is implemented by concatenating a downsampling unit ($\boldsymbol{D}$), a symmetric fully connected (SFC) layer ($\boldsymbol{\Phi}$) and an upsampling unit ($\boldsymbol{D}^T$), as depicted in model 2 of Fig. 3.

According to the above developed model 1 (for $\boldsymbol{Z}^{k+1}$) and model 2 (for $\boldsymbol{Y}_H^{k+1}$), the deep unfolding network for implementing the K-stage Algorithm 2 is deployed in Fig. 2. According to the philosophy of the CODE theory [30], we only need to have a simple rough solution to be used in the $\boldsymbol{Q}$-quadratic-norm regularization. So, following some typical settings [29], [57], we set the level of $K := 4$ in COS2A to avoid sophisticated network architecture. In Fig. 2, the kth stage ($1 < k < K$) can be understood as the preparation of the images "$\boldsymbol{Y}_H^k - \boldsymbol{U}^k$" [see (8)] and "$2\boldsymbol{D}^T\boldsymbol{Y}_S + \rho(\boldsymbol{Z}^{k+1} + \boldsymbol{U}^k)$" [see (9)], the processing of the prepared images using model 1 [see (8)] and model 2 [see (9)], as well as the update of the scaled dual variable $\boldsymbol{U}^k$ (see Algorithm 2). Besides, stage 1 has an additional spectral upsampling block serving as the initialization of $\boldsymbol{Y}_H^0$ (see Algorithm 2), while the last stage (i.e., Stage K) is greatly simplified as it just needs to return the target image $\boldsymbol{Z}^K$, as illustrated in Fig. 2. Therefore, we are able to obtain $\boldsymbol{Y}_{\mathrm{DE}}$ through the interpretable deep network designed above.

As the regularizer $\|\cdot\|_{\boldsymbol{Q}}$ is interpreted in the CODE theory as a simultaneous feature extraction and fusion operator [30, eq. (3)], the design of $\boldsymbol{Q}$ aims to extract useful information from $\boldsymbol{Y}_{\mathrm{DE}}$ and fuses the information into the target image $\boldsymbol{Y}_H$. The desired information from $\boldsymbol{Y}_{\mathrm{DE}}$ is obviously its spectral shape. To utilize the spectral information of $\boldsymbol{Y}_{\mathrm{DE}}$, there are two potential approaches, corresponding to the pixel- and region-level spectral information usages. Since the deep unfolding criterion (5) accepts the multiresolution input $\boldsymbol{Y}_S$ (including the low-/medium-resolution bands), making $\boldsymbol{Y}_{\mathrm{DE}}$ appears to be somewhat blurred. That said, a pixel-level approach would not be effective, and thus, we apply the region-level approach that forces neighboring pixels/spectra to cooperate with each

other. Specifically, an $r \times r$ region integration (judiciously modeled by some blurring kernel $\boldsymbol{B}$) is first performed on $\boldsymbol{Y}_{\mathrm{DE}}$, before it is used to $\boldsymbol{Q}$-regularize the ill-posed COS2A problem, thereby leading to

$$\boldsymbol{Q} \triangleq \boldsymbol{B}\boldsymbol{B}^T \otimes \boldsymbol{I}_M \in \mathbb{R}^{(ML)\times(ML)} \tag{10}$$

where $\otimes$ denotes the Kronecker product. Also, as the blurring effect comes from the low-/medium-resolution input bands of $\boldsymbol{Y}_S$ (the 60-m bands has a blurring factor 6, while the 20-m bands has a blurring factor 2), it is natural to model the kernel to have a blurring factor $r := 2$ as Sentinel-2 is dominated by the 20-m bands (there are six 20-m bands, while there are just two 60-m bands). Moreover, as those pixels in the 20-m bands (resp., 60-m bands) are simply copied four times (resp., 36 times) as mentioned in Section II-A, this amounts to the so-called uniform blurring, which has the mathematical form of $\boldsymbol{B} \triangleq \boldsymbol{I}_{L/r^2} \otimes (\mathbf{1}_{r^2}/r^2) \in \mathbb{R}^{L\times(L/r^2)}$ [33], where $\boldsymbol{I}_n$ and $\mathbf{1}_n$ denote the $n \times n$ identity matrix and the n-dimensional all-one vector, respectively.

The remaining task is to present the implementation details about the spectral response matrix $\boldsymbol{D}$. We had ever tried to learn a universal $\boldsymbol{D}$ from sufficiently large amounts of real AVIRIS and Sentinel-2 data, but it turned out to be less effective probably due to the spatially invariant attributes. A theoretically better approach is hence to adaptively learn the spectral response for each data. To simplify this challenging task, we recall Theorem 1, which suggests that the estimated spectral response will be simply used in the dual spatial super-resolution stage. That said, we can just focus on learning the spectral response $\widetilde{\boldsymbol{D}}$ for those high-spatial-resolution bands (i.e., 10-m bands) in $\widetilde{\boldsymbol{Y}}_S$ [34]. To avoid the responses being concentrated in limited hyperspectral bands, we adopt the ridge regression [58], which promotes uniformly distributed solutions, thereby leading to the estimation criterion, i.e.,

$$\widetilde{\boldsymbol{D}} := \arg\min_{\boldsymbol{D}\geq\boldsymbol{0}} \|\boldsymbol{D}\boldsymbol{Y}_{\mathrm{DE}} - \widetilde{\boldsymbol{Y}}_S\|_F^2 + \eta\|\boldsymbol{D}\|_F^2 \tag{11}$$

where $\eta := 0.0001$ is an empirical setting and $\boldsymbol{Y}_{\mathrm{DE}}$ is used as a surrogate of the real AVIRIS data. Note that in practice, we do not have real AVIRIS data in the COS2A problem. For simulation, uniform downsampling already works well, so (11) will be used for real-data experiments to promote uniformly distributed downsampling solutions.

We have completed the design of the COS2A algorithm (i.e., Algorithm 1) based on the CODE theory [30] to solve the highly ill-posed COS2A super-resolution problem, allowing historical Sentinel-2 data being computationally converted into their corresponding high-standard AVIRIS counterpart. For the first time, the COS2A super-resolution is effectively achieved via the interpretable deep network (i.e., Algorithm 2) and the spectral–spatial duality (i.e., Theorem 1). COS2A has been achieved for the first time, and its effectiveness will be experimentally demonstrated next.

III. Experimental Results and Discussion

This section comprehensively evaluates the proposed COS2A algorithm (i.e., Algorithm 1), demonstrating its effectiveness. The experimental design and dataset description are provided in Section III-A, where one can find a data preparation protocol customized for the COS2A problem. As there is no prior COS2A method in existing literature, Section III-B begins by reporting the baseline preparation, followed by the discussion about the qualitative and quantitative testing results. In Section III-C, we demonstrate the real-world applicability of the proposed COS2A method, directly working on real Sentinel-2 data, thereby yielding high-fidelity hyperspectral reconstruction results that hold good resemblance to real AVIRIS data over the same scene. In Section III-D, we conduct in-depth investigations into COS2A, including the ablation study and generalization ability test.

A. Experimental Setting

Since prior works never consider the COS2A problem, we can not find an existing experimental protocol or open training dataset for performance evaluation. In view of this, we propose a simulation protocol for fair quantitative evaluation, as well as a real-data collection protocol for demonstrating the real-world applicability of our method.

We first collect 646 AVIRIS hyperspectral images [15], each with a 256×256 spatial size, available from the AVIRIS dataset offered in [59, Table 1], and the ratio of training/validation/testing is 15:1:1. After removing water-vapor absorption bands (i.e., bands 1–10, 104–116, 152–170, and 215–224) [20], [35] from the original 224 AVIRIS bands (covering wavelengths from about 400 to 2500 nm), the 172-band AVIRIS data $\boldsymbol{Y}_H$ are obtained and used in the experiments. To simulate the spectral and spatial characteristics of Sentinel-2 data, we first design a spectral response matrix $\boldsymbol{D} \in \mathbb{R}^{12\times172}$ that maps the AVIRIS spectra to the 12 Sentinel-2 bands (after excluding the low-resolution band 10, which basically records only the cirrus information [34]). Specifically, according to the central wavelength and the bandwidth of each Sentinel-2 band [34], [36], we can define its wavelength coverage and accordingly identify the subset of the AVIRIS bands corresponding to that Sentinel-2 band; as we did not find a public spectral response between AVIRIS and Sentinel-2 sensors, AVIRIS bands in the subset are averaged to form the simulated Sentinel-2 band. Then, among the 12 simulated Sentinel-2 bands, those with 10-m band indices (i.e., 2, 3, 4, and 8) bands are kept unchanged to simulate real high-resolution Sentinel-2 bands, while those with 20-m band indices (i.e., 5, 6, 7, 9, 11, and 12) and those with 60-m band indices (i.e., 1, and 10) are uniformly blurred by factors of 2 and 6, respectively, resulting in a simulated multiresolution Sentinel-2 image $\boldsymbol{Y}_S$. The above procedure provides AVIRIS/Sentinel-2 training data pairs ($\boldsymbol{Y}_H$ and $\boldsymbol{Y}_S$) with ground truth $\boldsymbol{Y}_H$ for quantitatively evaluating the testing results.

On the other hand, to demonstrate the real-world applicability of the proposed COS2A algorithm, we also try hard to identify spatially and temporally overlapping AVIRIS/Sentinel-2 data pairs so that the quality of the spectrally super-resolved Sentinel-2 data (obtained via Algorithm 1) can be evaluated based on the real AVIRIS data. However, as the flying trajectories of AVIRIS/Sentinel-2 sensors are not aligned, image rotation calibration is also needed as a preprocessing stage, besides addressing the resolution inconsistency between the

TABLE I

REAL AVIRIS AND SENTINEL-2 DATA PAIRS ARE DOWNLOADED FROM [15] AND [36], RESPECTIVELY. ALL SITES ARE LOCATED IN AMERICA, ALL DATA WERE ACQUIRED IN AUGUST 2019. SENTINEL-2 SPECTRAL RANGES ARE COVERED BY THE AVIRIS SENSORS, AND SENTINEL-2 IMAGE SIZES ARE PRESENTED WITH RESPECT TO THE 10-M BANDS

Site Name	Location	AVIRIS			Sentinel-2		
		GSD (m)	Spectral Range (μm)	Image Size	GSD (m)	Spectral Range (μm)	Image Size
Mill Fire 2	CA	16	0.4–2.5	8544×803	10 / 20 / 60	0.443–2.19	13670×1285
Western Sierra Foothills	CA	14.8	0.4–2.5	10486×855	10 / 20 / 60	0.443–2.19	15519×1265
Caltech	CA	16.1	0.4–2.5	9319×810	10 / 20 / 60	0.443–2.19	15004×1304
Williams Fire (Rose)	WA	16.1	0.4–2.5	9793×782	10 / 20 / 60	0.443–2.19	15767×1259

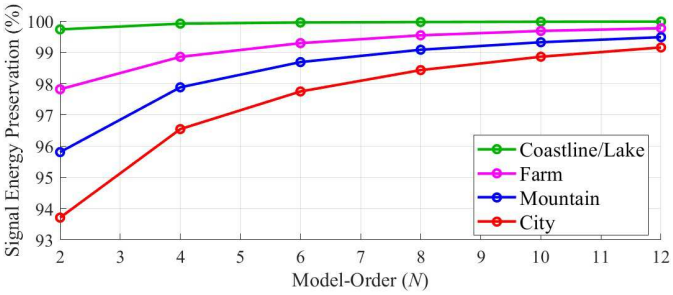


Fig. 4. Hyperspectral signal energy preservation for different model order N, tested on the 526 real AVIRIS data (see Section III-A) with results averaged for each land type. The signal energy preservation is defined as the ratio of low-rank image energy (i.e., $\|\boldsymbol{AS}\|_F^2$ with rank N) to the original image energy (i.e., $\|\boldsymbol{Y}_H\|_F^2$). Intriguingly, more smooth land types (e.g., lake) require smaller model orders to represent an image, while more detailed and intricate terrains (e.g., city) require higher model orders. The rank-8 approximation already well preserves about 99% hyperspectral signal energy, implying the validity of setting $N := 10$ in the proposed COS2A algorithm.

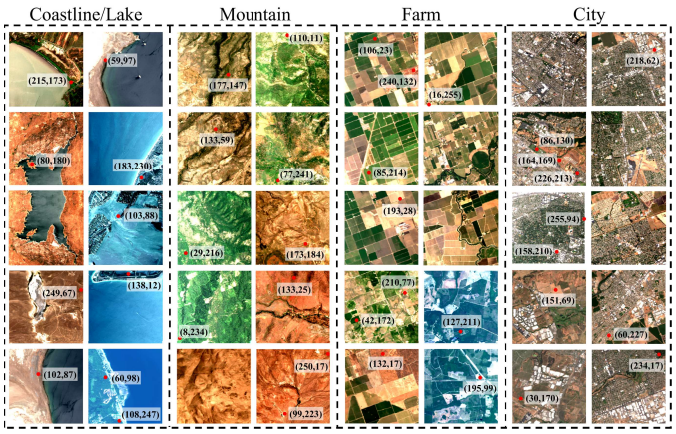


Fig. 5. Real AVIRIS images over the four representative land types, including coastline/lake, mountain, farm, and city. For each land type, we also select ten representative pixels (spectrally distinct pixels) as indicated by the red dots for the subsequent evaluation of spectral reconstruction quality (see Fig. 7).

two sensors. Specifically, we first download the AVIRIS Level-2 data (reflectance data) from [15] and Sentinel-2 Level-2A data from [36]. Then, the downloaded AVIRIS images are resampled to a 10-m spatial resolution using the benchmark ENVI software [60] in order to meet the highest resolution of Sentinel-2 data, and, as aforementioned, only 172 high-quality AVIRIS bands are used [20], [35]. Also, for Sentinel-2 data, the pixels in those 20-m bands (resp., 60-m bands) are already copied four times (resp., 36 times) when they are downloaded [36], so they are directly rotated to the direction aligning with the AVIRIS sensor trajectory; this orientation calibration can be done using the trajectory angle information provided in the AVIRIS metadata as well as the ENVI software [60]. We also conduct the image registration between AVIRIS/Sentinel-2 data using ENVI [60] to ensure accurate spatial alignment, thereby facilitating the learning of deep COS2A function. ENVI software is also used to remove irregular black borders and cloud-covered regions from the data, and the same cropping areas are applied to both AVIRIS/Sentinel-2 data to maintain their spatial consistency, thereby resulting in a total of 526 real-data pairs, each composed of a 256 × 256 × 172 AVIRIS image and a 256 × 256 × 12 Sentinel-2 image acquired over the same scene at the same time. The data are divided for training/validation/testing with the ratio of 8:1:1. Other related real-data information, including the acquisition sites, locations, ground sampling distances (GSD), spectral ranges, and spatial sizes, are summarized in Table I.

For the training setting of the proposed COS2A algorithm, the batch size is set to 8, and the deep unfolding network (see Fig. 2) is trained for 30 epochs. A total of 20 000 overlapped 64×64 patches are cropped for training and validation. Data augmentation methods, such as random horizontal/vertical flips and rotation, are applied during the training phase. The COS2A model is trained in an end-to-end manner by minimizing the outlier-robust ℓ_1 loss function [30], i.e., $\|\boldsymbol{Y}_H - \boldsymbol{Y}_{\text{DE}}\|_1$, where $\boldsymbol{Y}_H$ is the reference image and $\boldsymbol{Y}_{\text{DE}}$ is the network output during the training stage. The spectral response matrix $\boldsymbol{D}$ is known as discussed above, so it is applied for the simulation study, while it should be estimated via ridge regression for the real-data experiment. All other training configurations, including the optimizer, learning rate scheduling, and data augmentation strategies, are kept identical to those used for COS2A to ensure a fair comparison. All the experiments are conducted under MATLAB R2023b, Python 3.12.3/3.12.4, and PyTorch 2.6.0/2.7.0 on a computer equipped with an Intel Core i9-10900K CPU with 3.70-GHz speed, 46 46-GB of system RAM, and an NVIDIA GeForce RTX 3090 GPU.

For the parameter setting of the proposed COS2A algorithm, it is based on the derived duality and energy balancing, as explained next. For (α, β, λ), the parameters are induced from the spectral–spatial duality (see Appendix B), where the duality is implemented using the coupled-NMF (CNMF) algorithm as suggested by Theorem 1. A remarkable property of the adopted CNMF algorithm [33] is that its parameters have been successfully unified, with demonstrated robustness across highly diverse hyperspectral sensors, including ROSIS, HYDICE, as well as the target AVIRIS sensor [33]. Therefore, we simply follow the settings of $(1/2)(\alpha, \beta, \lambda) =$



Fig. 6. Spatial reconstruction results of (a) DAC, (b) MSFN, and (c) proposed COS2A algorithm over coastline/lake, mountain, farm, and city. All the images used are displayed in the true-color composition [B23(r)-B12(g)-B5(b)].

$(0.001, 0.001, 1)$ in [33]. As for η, it is set as 0.0001 for energy balancing in the ridge regression (11), and this order of ridge penalty follows some empirical settings [61] to avoid deteriorating the data fidelity [i.e., the first term of (11)]. Besides, the model order N is empirically set as 10 [33], which yields good performance in general as it typically ensures the preservation of more than 99% hyperspectral signal energy (see Fig. 4).

B. Qualitative and Quantitative Analysis

As this is the first work that achieves the COS2A function, which computationally upgrades the 12-band multiresolution Sentinel-2 data to 172-band AVIRIS-level hyperspectral data,

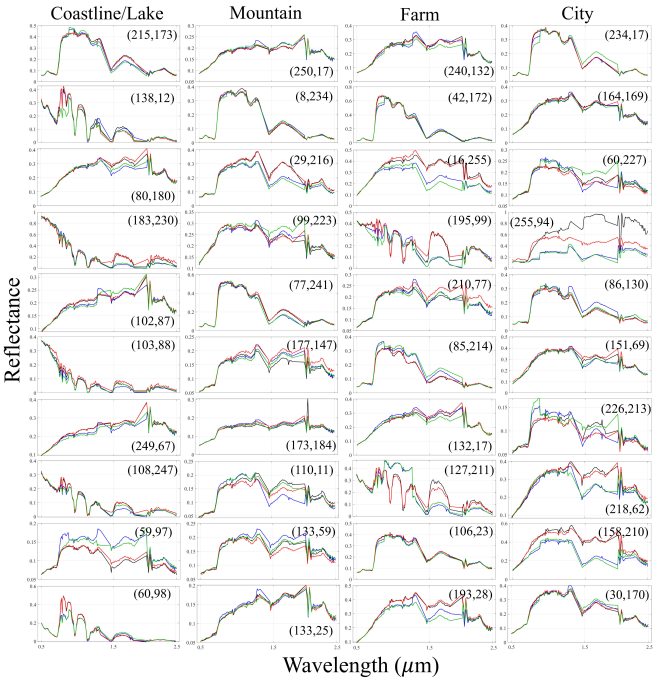


Fig. 7. Spectral reconstruction results of DAC, MSFN, and proposed COS2A algorithm over coastline/lake, mountain, farm, and city. The DAC-reconstructed, MSFN-reconstructed, and COS2A-reconstructed hyperspectral pixels are displayed as blue, green, and red curves, respectively, while the real AVIRIS pixels are displayed as black curves. The locations of the selected pixels are marked as red dots in Fig. 5.

TABLE II

QUANTITATIVE EVALUATION OF THE COS2A PERFORMANCE. THE BOLDFACED NUMBERS INDICATE THE BEST PERFORMANCES (I.E., THE LARGEST PSNR/SSIM OR THE SMALLEST SAM/RMSE)

	Methods	PSNR (↑)	SAM (↓)	RMSE (↓)	SSIM (↑)	Time (sec.)
Coastline/ Lake	DAC	27.6972	8.4138	0.0264	0.9285	0.0665
	MSFN	28.5163	7.5827	0.0257	0.9464	1.3051
	COS2A	**35.8897**	**2.4539**	**0.0080**	**0.9704**	15.3353
Mountain	DAC	28.2909	4.9266	0.0180	0.9317	0.0659
	MSFN	29.2526	4.0895	0.0148	0.9427	1.3025
	COS2A	**33.2310**	**2.0312**	**0.0092**	**0.9512**	14.5959
Farm	DAC	27.0976	9.0505	0.0427	0.8896	0.0654
	MSFN	28.0107	8.9927	0.0411	0.9034	1.2987
	COS2A	**35.0685**	**2.1460**	**0.0113**	**0.9527**	13.9210
City	DAC	28.0807	8.4681	0.0408	0.8230	0.0676
	MSFN	28.2156	6.2153	0.0341	0.8453	1.3007
	COS2A	**35.8443**	**3.3838**	**0.0175**	**0.9209**	14.6907
Average	DAC	27.7916	7.7147	0.0320	0.8932	0.0664
	MSFN	28.4988	6.7201	0.0289	0.9095	1.3018
	COS2A	**35.0084**	**2.5037**	**0.0115**	**0.9488**	14.6357

we have to design some preliminary baselines for comparison. As reported in Section I, almost all the related works focus on reconstructing 31-band hyperspectral images from three-band RGB images. Thus, it is natural to introduce the divide-and-conquer (DAC) algorithm [62] to prepare for the baseline. Specifically, as COS2A is a 12-to-172 super-resolution function, we divide the function into four parallel blocks, each conducting a 3-to-43 spectral super-resolution task. However, as most related works are for 3-to-31 super-resolution, they should be augmented with an additional fully connected layer

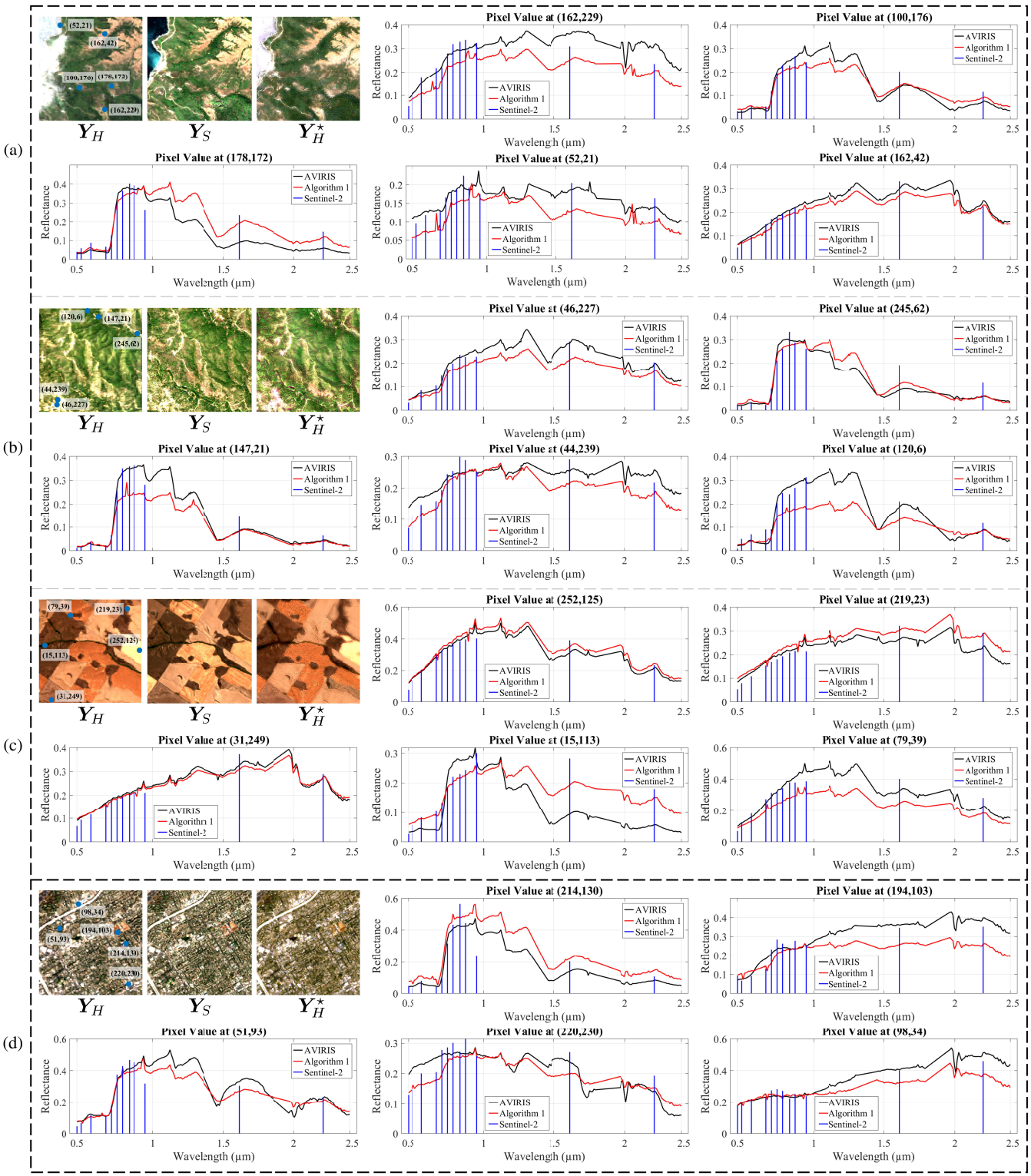


Fig. 8. Real-data testing results over (a) coastline, (b) mountain, (c) farm, and (d) city. For each land type, the real AVIRIS image $\boldsymbol{Y}_H$, its counterpart real Sentinel-2 image $\boldsymbol{Y}_S$, and the reconstructed AVIRIS image $\boldsymbol{Y}_H^\star$ (returned by Algorithm 1) are displayed for visual comparison, where both ($\boldsymbol{Y}_H$ and $\boldsymbol{Y}_H^\star$) use true-color composition [B23(r)-B12(g)-B5(b)]. The Sentinel-2 image $\boldsymbol{Y}_S$ also uses the true-color composition [B4(r)-B3(g)-B2(b)]. For each land type, we also randomly select five representative pixels with high spectral diversity and display their Sentinel-2 pixels (as blue pulses), real AVIRIS pixels (as black curves), and reconstructed AVIRIS pixels (as red curves).

to convert 31 feature maps into 43 ones. After each of the four blocks returns 43 features, they are aggregated into a 172-band hyperspectral image, which will be used as the baseline for our COS2A function. Each block here can be implemented using any CAVE-level method, and here, we opt to use the seminal MST algorithm [24], which won the

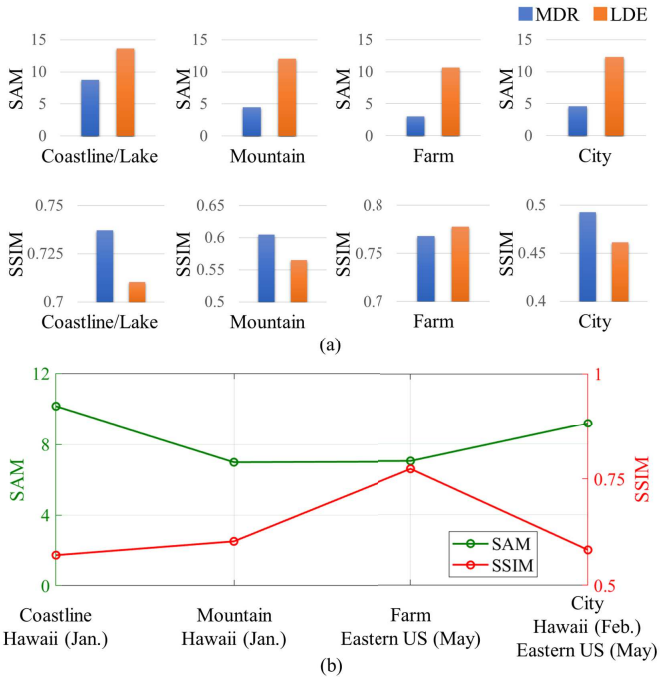


Fig. 9. (a) Comparison between the MDR (11) and the LDE of $\widetilde{\boldsymbol{D}}$ across different scenarios. (b) COS2A performances across more diverse spatial areas and seasons, demonstrating the robustness under deviated spatiotemporal conditions. SAM is the key measure for spectral structure similarity, while SSIM is the key measure for spatial structure similarity.

champion in the NTIRE 2022 spectral reconstruction challenge [63]. Note that although we have ever tried to directly modify the MST algorithm to fit the COS2A problem, it did not come up with an acceptable result before adopting the DAC strategy (even after nearly three-week training period). To be more specific, the performance of the seminal MST algorithm starts oscillating after epoch 30 for the challenging 12-to-172 spectral super-resolution mission, and higher epoch values even show some performance degradation compared to DAC. These facts motivate us to develop COS2A using novel theory, such as the spectral–spatial duality theory in Theorem 1. The reason is that even for such an outstanding MST algorithm [24], its configuration cannot handle the highly challenging COS2A mission, and that is why we have to introduce the DAC strategy to divide the COS2A problem into some simpler manageable subproblems. The above DAC configuration is trained over 30 epochs for each branch, enabling each branch to specialize in a specific and manageable portion of the spectrum learning, thereby effectively completing the DAC baseline design with MST (denoted as DAC for simplicity). Following a similar adaptation strategy described above, we also modify a very recent spectral super-resolution method [25] for the 12-to-172 super-resolution task. This baseline is denoted as the MSFN. The quantitative experimental results are summarized in Table II, as will be discussed later.

For effectively evaluating the similarity between COS2A-generated $\boldsymbol{Y}_H^{\star}$ and the real AVIRIS image $\boldsymbol{Y}_H$, we perform illumination calibration between the AVIRIS image and Sentinel-2 data (see Section III-C). We demonstrate the generalization ability of the COS2A across diverse scenes by categorizing the 40 testing data into four representative land types, namely, coastline/lake, mountain, farm, and city. Each land type includes ten data, and images for these four land types are displayed in Fig. 5. The visual reconstruction performance is presented in Fig. 6.

The proposed COS2A algorithm, DAC, and MSFN all exhibit outstanding spatial reconstruction results across most of the 40 data. However, some noticeable distortions are observed in the results of DAC. For example, in the coastline/lake type, the waterbody region in the first image of the left column shows a serious color deviation. Similarly, for the mountain type, the second image of the right column suffers from color inconsistency in the right half region. For the farm type, the bright regions in the fourth and fifth images of the right column are also discolored. In contrast, the proposed COS2A algorithm not only preserves fine spatial details across all the land types but also maintains color fidelity more consistently, alluding the spectral fidelity of the COS2A-reconstructed images as will be further validated next.

Since we test 40 images in total, each with 65 536 pixels, it is impractical to display all the pixel curves due to the page limitation. Thus, the ten images in each land type are jointly fed into the successive projection algorithm [64] to collaboratively identify ten representative pixels (i.e., spectrally distinct pixels) for each land type, as displayed in Fig. 5. The spectral reconstruction results for these selected pixels are collectively presented in Fig. 7. For the coastline/lake area, the pixel at (108, 247) demonstrates that the COS2A-reconstructed hyperspectral pixel highly overlaps with the real AVIRIS pixel, especially capturing the three continuous appeared peaks in the range of 0.7–0.95 μm. Remarkably, even for a more challenging pixel such as (59, 97), where the tested methods show noticeable deviations, the reconstructed pixels from these methods still preserve the overall spectral shape and remain similar to the real AVIRIS pixel. In the mountain area, the pixels at (99, 223) and (133, 25) show a high degree of alignment between the COS2A-reconstructed pixel and the AVIRIS one. For other pixels, such as (133, 59) and (117, 147), while slight deviations are observed, COS2A is able to preserve the overall spectral shape. In particular, the tested methods are able to capture the sharp oscillation patterns in the NIR spectral range. For the farm area, the pixel at (127, 211) appears to be a more challenging case for spectral reconstruction. Nevertheless, COS2A still generates a spectral curve that more closely follows the ground-truth one than DAC, highlighting its robustness under challenging conditions. The pixel at (16, 255) demonstrates that both methods achieve accurate spectral shapes, but COS2A again shows a much closer match to the real AVIRIS pixel, indicating its superior spectral fidelity. For the city area, which is generally considered a more complex area for spectral reconstruction due to its heterogeneous structures. We observe that the pixels, such as (255, 94) and (158, 210), exhibit more distinctive spectral characteristics. Even under these challenging conditions, COS2A consistently reconstructs spectra that are closer to the real AVIRIS pixels than DAC while still preserving a high spectral similarity. In addition, COS2A achieves excellent reconstruction results at pixel (30, 170), further demonstrating its effectiveness in urban areas. Overall, COS2A-reconstructed

pixels demonstrate highly consistent spectral shapes with real AVIRIS pixels (see Fig. 7), suggesting that it does effectively capture the fine characteristics of AVIRIS pixel features, hence leading to faithful spectral reconstruction results.

To comprehensively and fairly evaluate the performance of our method, we need to quantitatively analyze the results. Specifically, we adopt peak SNR (PSNR) [39], spectral angle mapper (SAM) [42], root-mean-squared error (RMSE) [59], and structural similarity (SSIM) [30] for quantitative evaluation. The results averaged over the testing data are summarized in Table II with the best performance of each scenario marked by a boldfaced number, where one can see that the spatial reconstruction quality achieved by COS2A is much superior to DAC and MSFN as indicated by the PSNR values. As for the SAM index, which is the most important measure for the spectral shape preservation, there is still a significant performance gap between COS2A and other methods (DAC and MSFN), where the significantly leading performance of COS2A (by several degrees) demonstrates the difficulty of the 12-to-172 spectral super-resolution task (no existing method can get it well done). MSFN performs the second best for the spectral reconstruction as indicated by SAM. Although DAC has reasonable spectral reconstruction results (less than 10°) [43], the COS2A-generated hyperspectral pixels show much better resemblance to the real AVIRIS hyperspectral pixels, as quantitatively validated in Table II. This well echoes the qualitative analysis in Fig. 7, although COS2A takes a longer (but reasonable) time to achieve so. RMSE is to measure global reconstruction quality. Given the outstanding spatial and spectral reconstruction quality (see PSNR and SAM indices in Table II), it is not surprising that COS2A achieves very low global reconstruction error as indicated by RMSE. Finally, as can be seen from the superior SSIM performance (see Table II), the COS2A-generated images do preserve outstanding structural similarity to the real AVIRIS images. This can also be observed by comparing the structural properties between the images displayed in Figs. 5 and 6(c).

Table II shows that the computational time of COS2A is the cost of its superior super-resolution performance. This echoes the derived quadratic complexity with respect to the number of spectral bands M (see Section II-C). Specifically, COS2A achieves the reconstruction of $M = 172$ AVIRIS-level hyperspectral bands for the first time, while such a large value of M (together with the quadratic complexity in M) explains the unsatisfactory computational efficiency. However, given the algorithmic breakthrough of achieving $M = 172$, the second-scale algorithmic computational time is reasonable and practically applicable, though the efficiency can be further improved in the future.

C. Real-Data Evaluation

This section aims to demonstrate the real-world applicability of the proposed COS2A algorithm (i.e., Algorithm 1). Specifically, by directly feeding real Sentinel-2 data $\boldsymbol{Y}_S$ into the algorithm, we compare the algorithmic output $\boldsymbol{Y}_H^\star$ with the real AVIRIS counterpart data, thereby showing their high spatial and spectral resemblances. The preparation of the counterpart AVIRIS data has been reported in Section III-A. The other training strategy also follows the same settings as described in Section III-A. To effectively evaluate the similarity between $\boldsymbol{Y}_H^\star$ (generated from real Sentinel-2 data $\boldsymbol{Y}_S$) and the real AVIRIS data, we perform illumination calibration between the real AVIRIS and real Sentinel-2 data. Mathematically, the pixel $\boldsymbol{a} \in \mathbb{R}^{172}$ in a real AVIRIS image is calibrated as "$\gamma^\star \boldsymbol{a}$" to fit its counterpart Sentinel-2 pixel $\boldsymbol{s} \in \mathbb{R}^{12}$ for some calibration scaler $\gamma^\star \geq 0$, defined as

$$\gamma^\star \triangleq \arg\min_{\gamma \geq 0} \ \frac{1}{2}\|\gamma \widetilde{\boldsymbol{a}} - \boldsymbol{s}\|_2^2$$

where $\widetilde{\boldsymbol{a}} \in \mathbb{R}^{12}$ (a subvector of $\boldsymbol{a}$) is formed by the 12 AVIRIS bands whose central wavelengths are closest to those of the 12 Sentinel-2 bands.

The calibrated real AVIRIS hyperspectral pixels (i.e., $\gamma^\star \boldsymbol{a}$) are displayed as black curves serving as the ground-truth pixels (that can be understood as the desired optimal outcomes of the proposed COS2A algorithm), as can be seen from Fig. 8. Their corresponding COS2A-generated pixels in $\boldsymbol{Y}_H^\star$ (returned by Algorithm 1) are then displayed as red curves in Fig. 8. Also, their counterpart real Sentinel-2 pixels (i.e., $\boldsymbol{s}$) are displayed as the discrete blue pulses in Fig. 8. We comprehensively investigate four representative land types (i.e., coastline, mountain, farm, and city). For each land type, the real AVIRIS image $\boldsymbol{Y}_H$, its counterpart real Sentinel-2 image $\boldsymbol{Y}_S$, and the reconstructed AVIRIS image $\boldsymbol{Y}_H^\star$ (returned by Algorithm 1) are also displayed in Fig. 8 for visual comparison. For each land type, we also randomly select five representative pixels with high spectral diversity and display their multispectral/hyperspectral curves in Fig. 8.

For the coastline area [see Fig. 8(a)], one can see that the spatial texture of $\boldsymbol{Y}_H^\star$ does hold high resemblance to that of $\boldsymbol{Y}_H$. It is remarkable that although the coastal area of $\boldsymbol{Y}_S$ (the upper left corner) has some color distortion, our proposed COS2A algorithm is able to automatically calibrate the color, thereby making $\boldsymbol{Y}_H^\star$ quite similar to $\boldsymbol{Y}_H$. One can also see that the real and reconstructed hyperspectral pixels (i.e., the black and red curves) are very similar. For the pixel at (178,172), the red curve returned by our COS2A algorithm even better fits the Sentinel-2 multispectral pixel when compared to the black curve, probably because the acquisition of AVIRIS data is somewhat affected by the illumination condition. For the mountain area [see Fig. 8(b)], the spatial details of $\boldsymbol{Y}_H^\star$ again hold high resemblance to the real AVIRIS data $\boldsymbol{Y}_H$, and COS2A-generated hyperspectral pixels are also well reconstructed from $\boldsymbol{Y}_S$. For the farm area [see Fig. 8(c)], even if $\boldsymbol{Y}_S$ has obvious illumination deviation from $\boldsymbol{Y}_H$, the super-resolved $\boldsymbol{Y}_H^\star$ very well preserves the color and structure of the real AVIRIS data $\boldsymbol{Y}_H$, preserving a clear boundary between different farmlands. The reconstructed hyperspectral pixels (red curves) also exhibit high fidelity compared to the real pixels (black curves). Remarkably, even the Sentinel-2 pixel indexed by (15,113) is quite deviated in the NIR region, the proposed COS2A algorithm can still return a hyperspectral pixel with very similar spectral shape compared to the real AVIRIS pixel in the NIR spectral range. For the city area [see Fig. 8(d)], the fine structural information in $\boldsymbol{Y}_S$ has been well fused into the $\boldsymbol{Y}_H^\star$, exhibiting consistent and faithful

spatial reconstruction performance of the COS2A algorithm. The spectral super-resolution is also well achieved with highly aligned real and reconstructed AVIRIS hyperspectral pixels. Therefore, we have proven that the proposed COS2A algorithm is able to yield high-fidelity hyperspectral reconstruction results that hold good resemblance to real AVIRIS data.

D. Discussion

Although the strong super-resolution results of COS2A would already imply the reliability of the proposed ridge regression (11), it is further validated in a more straightforward manner. Specifically, rather than using ridge regression, we directly attempt to learn a universal $\widetilde{\boldsymbol{D}}$ from the real dataset using deep regression (with each Sentinel-2 band associated with an $1 \times 1 \times 172$ convolution kernel), where the dataset contains 526 pairs of real AVIRIS image and Sentinel-2 image, as described in Section III-A. Then, the learning-based estimate of $\widetilde{\boldsymbol{D}}$ is plugged into the COS2A flow (see Fig. 1), replacing the ridge regression one obtained by (11). The comparison results are summarized in Fig. 9(a). One can observe that the mathematics-driven regression (MDR) (11) does yield much stronger COS2A reconstruction compared to the above learning-driven estimation (LDE), as validated across all the different scenarios covering four representative land types [see Fig. 9(a)]. Learning a universal spectral response turned out to be less effective in terms of SAM and SSIM metrics [see Fig. 9(a)], probably due to the spatially invariant attributes across different scenes. Therefore, a theoretically more reliable approach is to learn the spectral response adaptively for each data using (11), as suggested in Algorithm 1.

To further prove the generalizability of our COS2A algorithm, in addition to the Western USA region (see Table I), we further investigate the westernmost (i.e., Hawaii, USA) and easternmost (i.e., the Eastern USA) areas within the scope of openly available AVIRIS datasets. Specifically, we select coastline, mountain, and city in the Hawaii region, and farm and city in the Eastern USA region, with 7–10 samples for each case, to better reflect the regional diversity. The above selection intentionally included island and continental regions, aiming to have different atmospheric conditions. These data are also across different months for reflecting temporal diversity. The averaged performances are summarized in Fig. 9(b). For real data, an SAM value below ten indicates a good spectral structure similarity [43, Table V], whereas an SSIM value greater than 0.5 generally meets human vision perception [65, Sec. I]. According to Fig. 9(b), the results show good performances across different locations, land types, and seasons. Overall, these results highlight the robustness of the COS2A algorithm across diverse spatiotemporal conditions, proving its generalizability and potential applicability to a wide range of real-world scenarios.

On the other hand, in (3), we used the low-rank model $\boldsymbol{Y}_H = \boldsymbol{AS}$ with model order $N := 10$ for typical hyperspectral images [40], [41], [42]. To see the rationale, we show the hyperspectral signal energy preservation for different orders N in Fig. 4, where the results are obtained from the 526 real AVIRIS data (described in Section III-A) and averaged for each land type. Remarkably, approximating the hyperspectral data using a rank-3 matrix can already preserve up to 90% signal energy, and the rank-8 approximation could already preserve almost 99% signal energy for all the land types (see Fig. 4). The preserved energy covers not only the linear low-rank signal but also some nonlinear effects. By setting an even higher model order of $N := 10$, the proposed COS2A algorithm has more degrees of freedom to address more nonlinear effects, as a commonly used strategy to tackle the potential model imperfection [33, Sec. IV.C].

IV. Conclusion and Future Works

We have proposed the COS2A algorithm (i.e., Algorithm 1) based on a lightweight deep unfolding network (see Algorithm 2) and the spectral–spatial duality (see Theorem 1), thereby allowing Sentinel-2 multiresolution data to be computationally upgraded to AVIRIS-level hyperspectral data with uniformly high resolution (i.e., 10 m). We employ the CODE theory to mitigate the burden of the DE, thereby yielding pleasant COS2A results without sophisticated network architecture or laborious data collection procedure. Theories developed in this article are not only interpretable but also can be used to design critical spectral super-resolution algorithms for computationally generating other popular satellite images in the future. Experiments demonstrate the superiority and effectiveness of our proposed COS2A theory over various land cover types. In the future, remote sensing researchers could also use our COS2A technique to achieve hyperspectral missions (especially those requiring strong remote substances/objects identifiability) over regions covered by the Sentinel-2 satellite. Quantum-based real-time software implementation (e.g., by quantum generative models [35]) and energy-efficient hardware implementation of the COS2A algorithm also deserve future investigations to meet the edge-computing era.

Appendix

A. Proof of Theorem 1

To rigorously build the spectral–spatial duality, we first recall the well-known coupled-NMF spatial super-resolution problem in optical remote sensing [66], whose formally regularized version can be cast as [33, eq. (5)], which is copied here for the self-contained purpose

$$\min_{\boldsymbol{A},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\mathrm{CNMF}(\boldsymbol{A},\boldsymbol{S}) + \lambda_1 \mathrm{vol}(\boldsymbol{A}) + \lambda_2 \|\boldsymbol{S}\|_1 \tag{12}$$

where $\mathrm{CNMF}(\boldsymbol{A},\boldsymbol{S}) \triangleq \|\boldsymbol{Y}_H - \boldsymbol{ASB}\|_F^2 + \|\boldsymbol{Y}_M - \boldsymbol{DAS}\|_F^2$ is the so-called coupled-NMF formulation [66] used to spatially super-resolve the hyperspectral image $\boldsymbol{Y}_H$ by fusing the spatial details of the multispectral image $\boldsymbol{Y}_M$ into $\boldsymbol{Y}_H$. Therefore, it suffices to prove that (12) is mathematically equivalent to the target COS2A spectral super-resolution problem (4).

By comparing (4) and (12), one can first standardize the parameter setting as $\lambda_1 := 0.5\alpha$, $\lambda_2 := 0.5\beta$, and $\lambda := 2$ [see (4)]. Also, the high-resolution multispectral image $\boldsymbol{Y}_M$ for providing spatial details in the spatial super-resolution problem (12) can be set as the image composed of those 10-m Sentinel-2 bands, leading to $\boldsymbol{Y}_M := \widetilde{\boldsymbol{Y}}_S$, and thus, $\boldsymbol{D}$ in (12) is accordingly set as the spectral response $\widetilde{\boldsymbol{D}}$ used in

(4). Therefore, in order to prove the equivalence between (4) and (12), we only need to prove that the $\boldsymbol{Q}$-quadratic-norm term in (4) is equivalent to the first term in CNMF$(\boldsymbol{A},\boldsymbol{S})$, i.e., $\|\boldsymbol{Y}_H - \boldsymbol{ASB}\|_F^2 = \|\boldsymbol{AS} - \boldsymbol{Y}_{\rm DE}\|_{\boldsymbol{Q}}^2$. Furthermore, recalling from the discussion above (10), the $r \times r$ region integration (modeled by $\boldsymbol{B}$) is first performed on $\boldsymbol{Y}_{\rm DE}$ before it is used for $\boldsymbol{Q}$-regularization, i.e., $\boldsymbol{Y}_{\rm DE}\boldsymbol{B} = \boldsymbol{Y}_H$, implying that the equivalence between (4) and (12) can be built by proving

$$\|\boldsymbol{Y}_{\rm DE}\boldsymbol{B} - \boldsymbol{ASB}\|_F^2 = \|\boldsymbol{AS} - \boldsymbol{Y}_{\rm DE}\|_{\boldsymbol{Q}}^2 \tag{13}$$

which will be proven true below for building the desired spectral–spatial duality.

To prove (13), we notice that $\text{vec}(\boldsymbol{CEF}) = (\boldsymbol{F}^T \otimes \boldsymbol{C})\text{vec}(\boldsymbol{E})$ is true for any matrices $(\boldsymbol{C},\boldsymbol{E},\boldsymbol{F})$ with appropriate dimensionality, where $\text{vec}(\cdot)$ is the vectorization operator. Also, by recalling the definition of $\boldsymbol{B} \triangleq \boldsymbol{I}_{L/r^2} \otimes (\boldsymbol{1}_{r^2}/r^2) \in \mathbb{R}^{L\times(L/r^2)}$, we can build the following equalities, i.e.,

$$\begin{aligned}
&\|\boldsymbol{Y}_{\rm DE}\boldsymbol{B} - \boldsymbol{ASB}\|_F^2 \\
&= \left\|(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})\ \left[\boldsymbol{I}_{L/r^2} \otimes \left(\boldsymbol{1}_{r^2}/r^2\right)\right]\right\|_F^2 \\
&= \left\|\text{vec}\left\{\boldsymbol{I}_M\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})\ \left[\boldsymbol{I}_{L/r^2} \otimes \left(\boldsymbol{1}_{r^2}/r^2\right)\right]\right\}\right\|_2^2 \\
&= \left\|\left\{\left[\boldsymbol{I}_{L/r^2} \otimes \left(\boldsymbol{1}_{r^2}/r^2\right)\right]^T \otimes \boldsymbol{I}_M\right\}\text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})\right\|_2^2.
\end{aligned} \tag{14}$$

To simplify the derivation, let $\boldsymbol{v} \triangleq \text{vec}(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})$. We can then derive that

$$\begin{aligned}
&\|\boldsymbol{AS} - \boldsymbol{Y}_{\rm DE}\|_{\boldsymbol{Q}}^2 \\
&= \|\text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})\|_{\boldsymbol{Q}}^2 \\
&= \text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})^T\,\boldsymbol{Q}\text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS}) \\
&= \text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS})^T\,\left(\boldsymbol{BB}^T \otimes \boldsymbol{I}_M\right)\text{vec}\,(\boldsymbol{Y}_{\rm DE} - \boldsymbol{AS}) \\
&= \boldsymbol{v}^T\left(\boldsymbol{B}^T \otimes \boldsymbol{I}_M\right)^T\left(\boldsymbol{B}^T \otimes \boldsymbol{I}_M\right)\boldsymbol{v} \\
&= \left\|\left(\boldsymbol{B}^T \otimes \boldsymbol{I}_M\right)\boldsymbol{v}\right\|_2^2
\end{aligned} \tag{15}$$

where we have used the definition of $\boldsymbol{Q} = \boldsymbol{BB}^T \otimes \boldsymbol{I}_M \in \mathbb{R}^{(ML)\times(ML)}$ in (10). By the definitions of $(\boldsymbol{B},\boldsymbol{v})$, we further notice that (14) is equivalent to (15). In other words, $\|\boldsymbol{Y}_{\rm DE}\boldsymbol{B} - \boldsymbol{ASB}\|_F^2 = \|\boldsymbol{AS} - \boldsymbol{Y}_{\rm DE}\|_{\boldsymbol{Q}}^2$, which is exactly (13). Therefore, the spectral–spatial duality is proven, and the proof of Theorem 1 is completed.

As a side contribution of this proof, we notice that (4) is equivalent to [33, eq. (5)], and hence, (4) can be implemented by calling the coupled-NMF algorithm [33, Algorithm 1] with the input hyperspectral image set as $\boldsymbol{Y}_{\rm DE}\boldsymbol{B}$, input multispectral image set as $\widetilde{\boldsymbol{Y}}_S$, and input parameters empirically set as $(\lambda_1,\lambda_2) = 0.5(\alpha,\beta) = (0.001, 0.001)$. ■

B. Computational Complexity Analysis

The computational time of the proposed CODE-based COS2A algorithm is almost dominated by the CO part, whereas the DE part is almost executed in a near-real-time manner. According to Theorem 1, the CO part (4) is, however, executed by the coupled-NMF spatial super-resolution. Furthermore, according to Appendix A, the spatial super-resolution is implemented by the CO-based coupled-NMF algorithm proposed in [33], which is to solve (12) by alternatively optimizing $\boldsymbol{A}$ and $\boldsymbol{S}$.

For the complexity of $\boldsymbol{S}$, it is solved using [33, Algorithm 2], whose computational complexity can be proven to be $\mathcal{O}[(NL + (Nr^2)^2)\xi]$, where $\xi \triangleq \max\{M, M_m r^2, Nr^2\}$, as proved in [33, Lemma 1]. For the complexity of $\boldsymbol{A}$, it is solved using [33, Algorithm 3], whose computational complexity can be proven to be $\mathcal{O}[N^4M^3 + NM\xi']$, where $\xi' \triangleq \max\{ML/r^2, M_m L\}$, as proved in [33, Lemma 2]. In the COS2A problem, as $(N, r, M, M_m) = (10, 2, 172, 12)$, one can observe that $\xi = M$ and $\xi' = ML/r^2$. As a summary, the complexity for optimizing $\boldsymbol{S}$ is proven to be $\mathcal{O}[(NL + (Nr^2)^2)M]$, while the complexity for optimizing $\boldsymbol{A}$ is proven to be $\mathcal{O}[N^4M^3 + NM^2L/r^2]$.

Therefore, the overall computational complexity for COS2A has been derived as

$$\begin{aligned}
&\mathcal{O}\left[\left(NL + \left(Nr^2\right)^2\right)M + N^4M^3 + NM^2L/r^2\right] \\
&= \mathcal{O}\left[NML + \left(Nr^2\right)^2 M + N^4M^3 + \left(N/r^2\right)M^2L\right].
\end{aligned} \tag{16}$$

Although the complexity theoretically appears to scale as M^3, it is practically quadratic to the number of spectral bands M if one further considers the fact of $L \gg M \gg N > r$ in practice. Note that the image size L often affects the computational time most significantly. By (16), the complexity is linear to the scale of the image size L. ■

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References

[1] D. Phiri, M. Simwanda, S. Salekin, V. Nyirenda, Y. Murayama, and M. Ranagalage, "Sentinel-2 data for land cover/use mapping: A review," *Remote Sens.*, vol. 12, no. 14, pp. 1–35, Jul. 2020.

[2] J. Segarra, M. L. Buchaillot, J. L. Araus, and S. C. Kefauver, "Remote sensing for precision agriculture: Sentinel-2 improved features and applications," *Agronomy*, vol. 10, no. 5, pp. 1–18, May 2020.

[3] C.-H. Lin and J. M. Bioucas-Dias, "Nonnegative blind source separation for ill-conditioned mixtures via John Ellipsoid," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2209–2223, May 2021.

[4] C.-H. Lin, W.-K. Ma, W.-C. Li, C.-Y. Chi, and A. Ambikapathi, "Identifiability of the simplex volume minimization criterion for blind hyperspectral unmixing: The no-pure-pixel case," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5530–5546, Oct. 2015.

[5] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Apr. 2018.

[6] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[7] J.-T. Lin and C.-H. Lin, "SuperRPCA: A collaborative superpixel representation prior-aided RPCA for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532516.

[8] S.-S. Young, C.-H. Lin, and Z.-C. Leng, "Unsupervised abundance matrix reconstruction transformer-guided fractional attention mechanism for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9150–9164, May 2025.

[9] S. Galliani, C. Lanaras, D. Marmanis, E. Baltsavias, and K. Schindler, "Learned spectral super-resolution," 2017, *arXiv:1703.09470*.

[10] J. He et al., "Spectral super-resolution meets deep learning: Achievements and challenges," *Inf. Fusion*, vol. 97, pp. 1–22, Sep. 2023.

[11] R. Dian, Y. Liu, and S. Li, "Spectral super-resolution via deep low-rank tensor representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 5140–5150, Mar. 2025.

[12] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[13] J. Xue and B. Su, "Significant remote sensing vegetation indices: A review of developments and applications," *J. Sensors*, vol. 2017, pp. 1–17, May 2017.

[14] C.-H. Lin, M.-C. Chu, and P.-W. Tang, "CODE-MM: Convex deep mangrove mapping algorithm based on optical satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5620619.

[15] *AVIRIS Data Portal*. Accessed: Apr. 8, 2025. [Online]. Available: https://aviris.jpl.nasa.gov/dataportal/

[16] C.-H. Lin, S.-H. Huang, T.-H. Lin, and P. C. Wu, "Metasurface-empowered snapshot hyperspectral imaging with convex/deep (CODE) small-data learning theory," *Nature Commun.*, vol. 14, no. 1, pp. 1–10, Nov. 2023.

[17] R. Morales Ferre, J. Praks, G. Seco-Granados, and E. S. Lohan, "A feasibility study for signal-in-space design for LEO-PNT solutions with miniaturized satellites," *IEEE J. Miniaturization Air Space Syst.*, vol. 3, no. 4, pp. 171–183, Dec. 2022.

[18] C.-H. Lin and T.-H. Lin, "All-addition hyperspectral compressed sensing for metasurface-driven miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503215.

[19] X. Wang et al., "Advances in information processing and biological imaging using flat optics," *Nature Rev. Electr. Eng.*, vol. 1, no. 6, pp. 391–411, May 2024.

[20] C.-H. Lin and Y.-Y. Chen, "HyperQUEEN: Hyperspectral quantum deep network for image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511720.

[21] C.-H. Lin, T.-H. Lin, and J. Chanussot, "Quantum information-empowered graph neural network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537615.

[22] A. Liutkus, R. Badeau, and G. Richard, "Gaussian processes for under-determined source separation," *IEEE Trans. Signal Process.*, vol. 59, no. 7, pp. 3155–3167, Jul. 2011.

[23] C.-H. Lin and J.-T. Lin, "PRIME: Unsupervised multispectral unmixing using virtual quantum prism and convex geometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5615715.

[24] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 744–754.

[25] Y. Wu, R. Dian, and S. Li, "Multistage spatial–spectral fusion network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12736–12746, Jul. 2025.

[26] C. Wu, J. Li, R. Song, Y. Li, and Q. Du, "RepCPSI: Coordinate-preserving proximity spectral interaction network with reparameterization for lightweight spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508313.

[27] C.-H. Lin, T.-H. Lin, T.-H. Lin, and T.-H. Lin, "Fast reconstruction of hyperspectral image from its RGB counterpart using ADMM-adam theory," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Rome, Italy, Sep. 2022, pp. 1–5.

[28] B. Chen, L. Liu, C. Liu, Z. Zou, and Z. Shi, "Spectral-cascaded diffusion model for remote sensing image spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528414.

[29] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Deep unfolding network for spatiospectral image super-resolution," *IEEE Trans. Comput. Imag.*, vol. 8, pp. 28–40, 2022.

[30] C.-H. Lin, Y.-C. Lin, and P.-W. Tang, "ADMM-ADAM: A new inverse imaging framework blending the advantages of convex optimization and deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514616.

[31] C.-H. Lin and P.-W. Tang, "Inverse problem transform: Solving hyperspectral inpainting via deterministic compressed sensing," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[32] P.-W. Tang, C.-H. Lin, and Y. Liu, "Transformer-driven inverse problem transform for fast blind hyperspectral image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504414.

[33] C.-H. Lin, F. Ma, C.-Y. Chi, and C.-H. Hsieh, "A convex optimization-based coupled nonnegative matrix factorization algorithm for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1652–1667, Mar. 2018.

[34] C.-H. Lin and J. M. Bioucas-Dias, "An explicit and scene-adapted definition of convex self-similarity prior with application to unsupervised sentinel-2 super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3352–3365, May 2020.

[35] C.-H. Lin and S.-S. Young, "HyperKING: Quantum-classical generative adversarial networks for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5512119.

[36] *Sentinel-2 Google Earth Engine*. Accessed: Apr. 8, 2025. [Online]. Available: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUSS2SRHARMONIZED

[37] T.-H. Lin and C.-H. Lin, "Hyperspectral change detection using semi-supervised graph neural network and convex deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515818.

[38] C.-Y. Chi, W.-C. Li, and C.-H. Lin, *Convex Optimization for Signal Processing and Communications: From Fundamentals To Applications*. Boca Raton, FL, USA: CRC Press, 2017.

[39] C.-H. Lin, C.-Y. Hsieh, and J.-T. Lin, "CODE-IF: A convex/deep image fusion algorithm for efficient hyperspectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5617318.

[40] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[41] Y. Chen, Y. Guo, Y. Wang, D. Wang, C. Peng, and G. He, "Denoising of hyperspectral images using nonconvex low rank matrix approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5366–5380, Sep. 2017.

[42] C.-H. Lin and S.-S. Young, "Signal subspace identification for incomplete hyperspectral image with applications to various inverse problems," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512316.

[43] C.-H. Lin, C.-Y. Chi, Y.-H. Wang, and T.-H. Chan, "A fast hyperplane-based minimum-volume enclosing simplex algorithm for blind hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 64, no. 8, pp. 1946–1961, Apr. 2016.

[44] P. V. Giampouras, K. E. Themelis, A. A. Rontogiannis, and K. D. Koutroumbas, "Simultaneously sparse and low-rank abundance matrix estimation for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4775–4789, Aug. 2016.

[45] M. D. Craig, "Minimum-volume transforms for remotely sensed data," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 3, pp. 542–552, May 1994.

[46] J. M. P. Nascimento and J. M. Bioucas-Dias, "Hyperspectral unmixing based on mixtures of Dirichlet components," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 863–878, Mar. 2012.

[47] C.-H. Lin, A. Ambikapathi, W.-C. Li, and C.-Y. Chi, "On the end-member identifiability of Craig's criterion for hyperspectral unmixing: A statistical analysis for three-source case," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Vancouver, BC, Canada, May 2013, pp. 2139–2143.

[48] L. Zhuang, C.-H. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[49] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[50] S. Cheol Park, M. Kyu Park, and M. Gi Kang, "Super-resolution image reconstruction: A technical overview," *IEEE Signal Process. Mag.*, vol. 20, no. 3, pp. 21–36, May 2003.

[51] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, Oct. 1999.

[52] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[53] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[54] N. Parikh and S. Boyd, "Proximal algorithms," *Found. Trend Optim.*, vol. 1, no. 3, pp. 127–239, Jan. 2013.

[55] K. Zhang, Y. Li, W. Zuo, L. Zhang, L. Van Gool, and R. Timofte, "Plug-and-Play image restoration with deep denoiser prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 10, pp. 6360–6376, Oct. 2022.

[56] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[57] J. Wang, Z. Shao, X. Huang, T. Lu, and R. Zhang, "A deep unfolding method for satellite super resolution," *IEEE Trans. Comput. Imag.*, vol. 8, pp. 933–944, 2022.

[58] G. C. McDonald, "Ridge regression," *Wiley Interdiscipl. Reviews: Comput. Statist.*, vol. 1, no. 1, pp. 93–100, Jul. 2009.

[59] C.-C. Hsu, C.-H. Lin, C.-H. Kao, and Y.-C. Lin, "DCSN: Deep compressed sensing network for efficient hyperspectral data transmission of miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7773–7789, Sep. 2021.
[60] *ENVI 5.2*. Accessed: Mar. 28, 2025. [Online]. Available: https://www.nv5geospatialsoftware.com/
[61] D. B. Huberman, B. J. Reich, and H. D. Bondell, "Nonparametric conditional density estimation in a deep learning framework for short-term forecasting," *Environ. Ecol. Statist.*, vol. 29, no. 4, pp. 677–704, Dec. 2022.
[62] R. Sedgewick and K. Wayne, *Algorithms*. Boston, MA, USA: Addison-Wesley, 2011.
[63] B. Arad et al., "NTIRE 2022 spectral recovery challenge and data set," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 863–881.
[64] S. Arora et al., "A practical algorithm for topic modeling with provable guarantees," 2012, *arXiv:1212.4777*.
[65] T.-C. Chang, S. S.-D. Xu, and S.-F. Su, "SSIM-based quality-on-demand energy-saving schemes for OLED displays," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 46, no. 5, pp. 623–635, May 2016.
[66] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.



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